

KHELIF ADEM ABDELBASSET

Green Hydrogen | Techno-Economic Modeling | PEMFC & Electrolysis | Process Engineering

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PROFESSIONAL SUMMARY

Renewable energy engineer specializing in green hydrogen systems, techno-economic modeling, and PEMFC multiphysics modeling, graduating top of class (18–19/20) from HNS-RE²SD under the Algerian-German TaqatHy+ collaboration with GIZ and the University of Rostock. Experienced across the full green hydrogen value chain — PEM/alkaline/SOEC electrolysis, seawater desalination, and PEMFC modeling — with hands-on exposure to industrial automation (Siemens S7 PLC systems) and hybrid renewable energy systems. Delivered pilot-scale green hydrogen and ammonia feasibility studies (HOMER Pro, Aspen Plus) with quantified LCOH, NPV, and IRR outcomes, and multi-scenario decarbonization roadmaps aligned with EU CBAM compliance.

EDUCATION

Master's Degree — Renewable Energies, Green Hydrogen Option

Higher National School of Renewable Energies, Environment & Sustainable Development (HNS-RE²SD), Algeria 2026

Grade: 18/20 — Algerian-German TaqatHy+ Green Hydrogen Collaboration, with GIZ and the University of Rostock

Thesis: “Techno-Economic Study of a Spatially Decoupled Green Hydrogen Production System via Renewable Energy and Seawater Desalination in Algeria”

- Optimized a 100%-renewable hybrid system in HOMER Pro (63.5 MWp PV, ~64.2 MW wind, 50 MW electrolyzer) with a DuPont WAVE desalination train, delivering LCOH \$3.69/kg, NPV \$44.5M, IRR 10.13%

Engineering Degree (Ingénieur d'État) — Renewable Energies, Green Hydrogen Option

Higher National School of Renewable Energies, Environment & Sustainable Development (HNS-RE²SD), Algeria 2026

Grade: 19/20

Thesis: “Physics-Based Modeling, Sensitivity Analysis and Optimization Study of Proton Exchange Membrane Fuel Cells”

- Developed a 1D multiphysics PEMFC model in MATLAB, validated against Ballard/Horizon data; Sobol sensitivity analysis and fmincon (SQP) optimization raised power density to 1.17 W/cm² (2.5× baseline)

Relevant Coursework: Power Electronics, Electrical Machines, Applied Thermodynamics, Scientific Computing & Programming, Renewable Energy Resource Assessment, Materials Science, Control Systems, Heat Transfer, Electrical Networks, Microprocessors & Microcontrollers, Renewable Energy Conversion, Thermofluids, PV & Wind Power Plant Design/Optimization, Hydrogen-Based Electricity Production, Energy Systems Modeling & Optimization, Solar Thermal Collector Technology, PV Cell Technology, Business Economics, Hydrogen Chemistry, Electrochemistry, Water Quality, Green Hydrogen Production/Storage/Transport & Distribution, Hydrogen Energy Chain, Electrochemical & Thermal Conversion of Hydrogen, Circular Economy, Industrial Applications of Green Hydrogen, Techno-Economic Optimization of Hybrid Systems, Project Management & Life-Cycle Assessment, Power-to-X Technologies, Green Hydrogen Systems Modeling, Hydrogen Risk & Safety, Hydrogen Market & Ecosystem, Energy Transition Legislation & Regulation, Technology Trends, Scientific Writing & Patents, CO₂ Capture & Green Hydrogen Production, AI for Renewable Energy & Green Hydrogen Systems

Bachelor in Electronics

Université de Batna 2, Batna, Algeria

2023

Graduation Project: “IoT Home” — ESP32-based smart home automation system

PROFESSIONAL EXPERIENCE

Renewable Energy Intern

Centre de Développement des Energies Renouvelables (CDER), Bouzareah, Algeria

February 2025

- Worked across CDER's solar, wind, and hybrid energy divisions, building foundational knowledge of renewable systems and rural electrification strategy

Automation & Digitalization Intern

Siemens, Digital Industries Division, Hydra, Algeria

July 2024

- Completed an immersion internship in automation and digitalization; gained hands-on experience with Siemens S7-1200 and S7-300 PLC systems

Research Observer Intern

Observed research projects on solar thermal systems, photovoltaic technologies, and energy-efficient equipment	
Key Projects & Research	
HyGreen Algeria — Hybrid Solar-Wind SOEC System for Green Hydrogen & Ammonia Co-Production	
Techno-Economic & Project Management Project	2026
- Led techno-economic feasibility and PM planning for a pilot-scale green ammonia plant (8 MWp PV + 2 MW wind + 1 MW Sunfire SOEC electrolysis at 82.3% efficiency, Casale 1.5 TPD Haber-Bosch with waste-heat recovery, ≥68% system efficiency), targeting LCOH ≤340 DZD/kg and LCOA ≤61,000 DZD/ton (25% below fossil ammonia) - Delivered CAPEX/OPEX modeling, supplier selection, and an 18-month EPCM framework for Sonatrach's Renewable Energy Division, coordinating regulators, Sonelgaz, and investors (EIB, GIZ) - LCA showed ≥7,500 t/yr CO₂ avoidance (95.6% vs. grey ammonia); established 22 KPIs across 5 pillars	
The Green Transition: Decarbonizing the Algerian Cement Industry	
Energy Transition, Legislation & Regulation Project	2026
- Designed a decarbonization roadmap for Algeria's cement sector (41 Mt/yr, 26.69 Mt CO₂/yr) for EU CBAM compliance, targeting carbon intensity <300 kg CO₂/t by 2040 (54% reduction) and >70% renewable share - Modeled three scenarios (150B–1.5T DZD), recommending a hybrid Green Hydrogen + CCUS strategy (80% capture, 346.98B DZD/yr) with a National Green Cement Task Force coordinating GICA, Sonelgaz, and investors (EIB, GIZ), aligned with EU CBAM and Algeria's Hydrogen Roadmap (40 TWh by 2040)	
Green Ammonia Production via Water Electrolysis — Aspen Plus Simulation	
Batna, Algeria	2026
Simulated a green ammonia production process in Aspen Plus, coupling a water electrolyzer with an air separation unit (N₂) and a Haber-Bosch synthesis loop; modeled mass/energy balances and separation/recycle streams to assess process feasibility	
Smartwatch for Healthcare Assistance with AI — ESP32-Based	
Boumerdes, Algeria	March 2023
Developed a wearable ESP32 health-monitoring smartwatch tracking heart rate, body temperature, and SpO₂, with AI-based real-time anomaly detection	
IoT Home — Bachelor's Graduation Project	
Batna, Algeria	May 2023
Built an ESP32-based smart home system for remote control/monitoring of lights, fans, and sensors via a mobile/web interface	

SKILLS

Soft Skills

- Technical & Scientific Writing, Investor Pitching, Stakeholder Governance
- Cross-Cultural Communication, Multilingual Presentation

Technical Skills

- Green Hydrogen Technology, Green Ammonia Synthesis, Techno-Economic Modeling, PEM Fuel Cell Systems, Hybrid Renewable Design, Electrolysis (PEM/Alkaline/SOEC)
- Programming & Simulation: C, C++, Python, MATLAB/Simulink, Modelica, Aspen Plus, TRNSYS, HOMER Pro, PVsyst, DuPont WAVE, CycleTempo
- Analysis Methods: Sobol GSA, fmincon (SQP), Monte Carlo Simulation, LCA, Exergy Analysis, Financial Modeling (NPV/IRR/LCOH), Data Analysis
- Project & Risk Management: EPCM Frameworks, WBS Planning, Risk Allocation, Supplier Selection, KPI Design
- Policy & Regulatory Analysis: EU CBAM Compliance, Renewable Energy Policy (PENREE), National Hydrogen Strategy, Circular Economy Strategy
- Design & Tools: Embedded Systems, Microcontroller Programming, IoT/Wireless Systems, SolidWorks, Blender, Proteus/PCB Design, LaTeX, Canva

LANGUAGES

Arabic — Native | French — Advanced | English — Proficient